Code No: 114CS **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, July/August - 2021
DESIGN AND ANALYSIS OF ALGORITHMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Write an algorithm to search a key k in a N*N matrix, in which each row is sorted is ascending order. What is the time complexity of your algorithm? [15]

2.a) Explain non recursive pre order tree traversal technique with example.
b) Explain how to improve the time complexity of union algorithm. [8+7]

3. Solve the following instance of 0/1 Knapsack problem using Dynamic programming.
n = 3; (W1, W2, W3) = (3, 5, 7);
(P1, P2, P3) = (3, 7, 12); M = 4. [15]

4.a) Explain with the example how to find a Hamiltonian cycle.
b) Explain how branch and bound technique is used to solve knapsack problem. [7+8]

5.a) Write the differences between NP complete and NP hard problems.
b) State and prove Cook's theorem. [6+9]

6. Derive the time complexity of strassen's matrix multiplication. [15]

7.a) Write an algorithm of AND/OR graph.
b) Differentiate between BFS and DFS. [9+6]

8. Explain about multistage graph with an example. [15]

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